

2026 Classical Mechanics Midterm.

1. Find two coupled 1st order ordinary differential equations from the 2nd order ordinary differential equation of the hypergeometric function.
 - (1) Find the corresponding Lagrangian to give the 2nd order hypergeometric equation via the Euler-Lagrange equation of motion.
 - (2) Performing the Legendre transformation, find the corresponding Hamiltonian and the Hamiltonian equation of motion.
 - (3) Through the canonical transformation, find the Hamilton-Jacobi equation for the hypergeometric equation.
2. Quantize the hypergeometric equation.
 - (1) Find the corresponding Schrodinger equation for the hypergeometric equation. Show that the semiclassical approximation ($\hbar \rightarrow 0$) reduces this Schrodinger equation to the above Hamilton-Jacobi equation. Can you find the eigenfunctions? You can use the Mathematica.
 - (2) Find the corresponding Heisenberg equation of motion for the hypergeometric equation. Can you introduce creation & annihilation operators to diagonalize the Hamiltonian operator? Construct the Hilbert space.
 - (3) Find the path integral formulation to show that it is a solution of the above Schrodinger equation. Can you perform the path integral explicitly, certainly possible in principle?